## #10 - Towards Autonomous Pose Estimation for Unseen Satellites

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Future space activities such as in-orbit construction, debris removal and far side lunar exploration require greatly improved autonomy. This increased autonomy is largely expected to be driven by a foundation of neural networks within space systems. Proximal pose estimation of non-cooperative targets is a key enabling technology, but with many challenges to address. In particular, (1) estimating the pose of previously unseen objects; (2) the compression of neural networks for deployment on limited space hardware; (3) the lack of available data for training these algorithms; and (4) ensuring the algorithms' robustness in the challenging space environment. This paper introduces the projects, both completed and ongoing, at EPFL CVLab to address these challenges. We demonstrate the pose estimation of a rocket body while training on images of household objects as well as compressing networks by a factor of 8 while maintaining state-of-the-art performance.